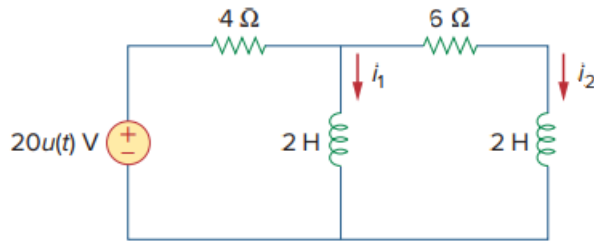


Problem

Obtain i_1 and i_2 for $t > 0$ in the circuit of Fig. 8.106.

Figure 8.106:



Step-by-step solution

Step 1 of 9

Refer to circuit diagram in Figure 8.106 in the text book.

It is known that,

$$u(t) = \begin{cases} 0 & \text{for } t < 0 \\ 1 & \text{for } t \geq 0 \end{cases}$$

At $t = 0^-$, the value of voltage source is 0 V. so, the current through the inductors at $t = 0^-$ are zero.

That is,

$$i_1(0^-) = 0 \text{ A}$$

$$i_2(0^-) = 0 \text{ A}$$

Step 2 of 9

Draw the circuit diagram at $t > 0$. The value of the voltage source is 20 V since the value of $u(t)$ at $t > 0$ is 1.

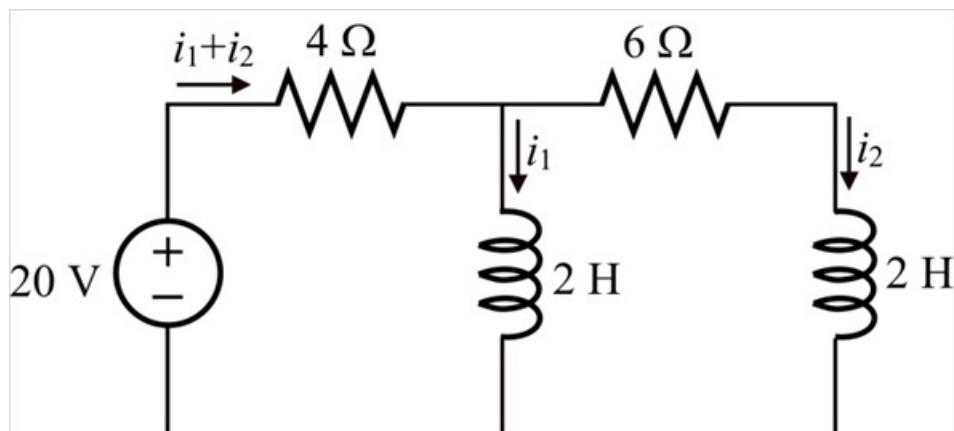


Figure 1

Step 3 of 9

The current through the inductor will not change instantaneously. Hence, at $t > 0$ the value of the currents $i_1(0^+)$ and $i_2(0^+)$ are 0 A .

Notice the current through the inductor is continuous. Therefore, the value of the currents $i_1(0)$ and $i_2(0)$ are 0 A .

Apply Kirchhoff's voltage law to left side loop.

$$-20 + 4(i_1 + i_2) + 2 \frac{di_1}{dt} = 0$$

$$-10 + 2i_1 + 2i_2 + \frac{di_1}{dt} = 0$$

$$2i_2 = 10 - 2i_1 - \frac{di_1}{dt}$$

$$i_2 = 5 - i_1 - 0.5 \frac{di_1}{dt} \dots\dots (1)$$

Step 4 of 9

Apply Kirchhoff's voltage law to right side loop.

$$-2 \frac{di_1}{dt} + 6i_2 + 2 \frac{di_2}{dt} = 0$$

$$-\frac{di_1}{dt} + 3i_2 + \frac{di_2}{dt} = 0$$

Substitute $5 - i_1 - 0.5 \frac{di_1}{dt}$ for i_2 .

$$-\frac{di_1}{dt} + 3\left(5 - i_1 - 0.5 \frac{di_1}{dt}\right) + \frac{d}{dt}\left(5 - i_1 - 0.5 \frac{di_1}{dt}\right) = 0$$

$$-\frac{di_1}{dt} + 15 - 3i_1 - 1.5 \frac{di_1}{dt} + 0 - \frac{di_1}{dt} - 0.5 \frac{d^2 i_1}{dt^2} = 0$$

$$-0.5 \frac{d^2 i_1}{dt^2} - 3.5 \frac{di_1}{dt} - 3i_1 = -15$$

$$\frac{d^2 i_1}{dt^2} + 7 \frac{di_1}{dt} + 6i_1 = 30$$

The characteristic equation of the circuit is,

$$s^2 + 7s + 6 = 0$$

$$(s + 6)(s + 1) = 0$$

The roots of the equation are -6 and -1 .

The roots are negative and real, hence the circuit exhibits over damped response.

The expression for current through the inductor is,

$$i_1(t) = i_1(\infty) + A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$i_1(t) = i_1(\infty) + A_1 e^{-t} + A_2 e^{-6t} \dots\dots (2)$$

Step 5 of 9

The circuit diagram at $t = \infty$ is shown in Figure 2.

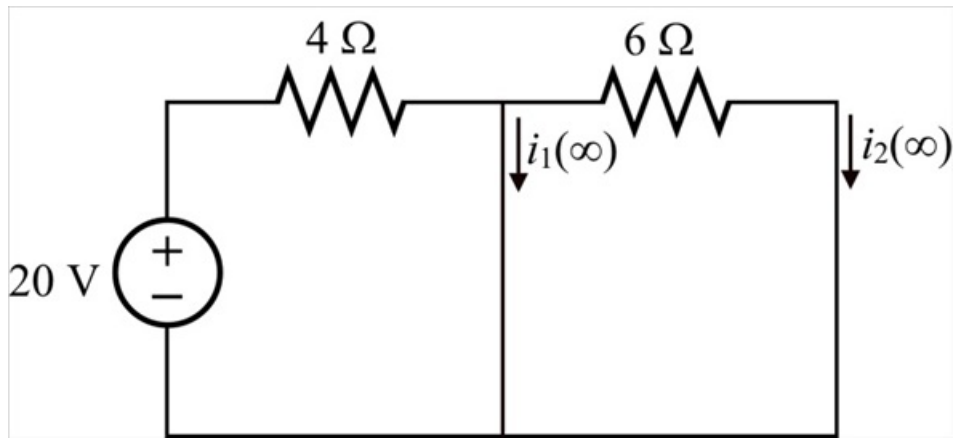


Figure 2

Step 6 of 9

Calculate the value of the current $i_1(\infty)$.

$$\begin{aligned} i_1(\infty) &= \frac{20}{4} \\ &= 5 \text{ A} \end{aligned}$$

Calculate the value of the current $i_2(\infty)$.

$$i_2(\infty) = 0 \text{ A}$$

Recall equation (2).

$$i_1(t) = i_1(\infty) + A_1 e^{-t} + A_2 e^{-6t}$$

Substitute 5 A for $i_1(\infty)$.

$$i_1(t) = 5 + A_1 e^{-t} + A_2 e^{-6t} \dots\dots (3)$$

Substitute 0 for t .

$$i_1(0) = 5 + A_1 e^{-(0)} + A_2 e^{-6(0)}$$

$$0 = 5 + A_1 + A_2$$

$$A_1 + A_2 = -5 \dots\dots (4)$$

Step 7 of 9

Recall equation (1).

$$i_2 = 5 - i_1 - 0.5 \frac{di_1}{dt}$$

Apply initial conditions.

$$i_2(0) = 5 - i_1(0) - 0.5 \frac{di_1(0)}{dt}$$

$$0 = 5 - 0 - 0.5 \frac{di_1(0)}{dt}$$

$$\frac{di_1(0)}{dt} = 10 \text{ A/s}$$

Differentiate equation (3) with respect to t .

$$\frac{di_1(t)}{dt} = 0 - A_1 e^{-t} - 6A_2 e^{-6t}$$

Substitute 0 for t .

$$\frac{di_1(0)}{dt} = -A_1 e^{-(0)} - 6A_2 e^{-6(0)}$$

$$10 = -A_1 - 6A_2 \quad \left(\text{since } \frac{di_1(0)}{dt} = 10 \right)$$

$$A_1 + 6A_2 = -10 \quad \dots\dots (5)$$

Step 8 of 9

Solve equations (4) and (5).

$$A_1 = -4$$

$$A_2 = -1$$

Recall equation (3).

$$i_1(t) = 5 + A_1 e^{-t} + A_2 e^{-6t}$$

Substitute -4 for A_1 and -1 for A_2 .

$$\begin{aligned} i_1(t) &= 5 + (-4)e^{-t} + (-1)e^{-6t} \\ &= 5 - 4e^{-t} - e^{-6t} \text{ A} \end{aligned}$$

Therefore, the current $i_1(t)$ for $t > 0$ is $\boxed{5 - 4e^{-t} - e^{-6t} \text{ A}}$.

Step 9 of 9

Recall equation (1).

$$i_2 = 5 - i_1 - 0.5 \frac{di_1}{dt}$$

Substitute $5 - 4e^{-t} - e^{-6t}$ A for $i_1(t)$.

$$\begin{aligned} i_2(t) &= 5 - [5 - 4e^{-t} - e^{-6t}] - 0.5 \frac{d}{dt} [5 - 4e^{-t} - e^{-6t}] \\ &= 5 - 5 + 4e^{-t} + e^{-6t} - (0.5)[0 - (4)(-1)e^{-t} - (-6)e^{-6t}] \\ &= 5 - 5 + 4e^{-t} + e^{-6t} - 2e^{-t} - 3e^{-6t} \\ &= 2e^{-t} - 2e^{-6t} \text{ A} \end{aligned}$$

Therefore, the current $i_2(t)$ for $t > 0$ is $\boxed{2e^{-t} - 2e^{-6t} \text{ A}}$.